**R13**

**Code No: 115EH**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech III Year I Semester Examinations, July/August - 2023**
**OPERATING SYSTEMS**
**(Common to CSE, IT)**

**Time: 3 hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

**PART - A**

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | Define system call. | [2] |
| b) | What are the OS features required for multiprogramming? | [3] |
| c) | Define scheduler. | [2] |
| d) | Differentiate preemptive and non-preemptive scheduling. | [3] |
| e) | How does the system detect thrashing? | [2] |
| f) | Define external fragmentation. | [3] |
| g) | What are the Access methods available? | [2] |
| h) | Differentiate between file and directory. | [3] |
| i) | Define deadlock. | [2] |
| j) | "If there is a cycle in the resource allocation graph, it may or may not be in deadlock state". Justify the statement. | [3] |

**PART - B**

**(50 Marks)**

| | | |
|---|---|---|
| 2.a) | What is operating system? Explain multiprogramming and time sharing systems. | |
| b) | What are the differences between loosely coupled and tightly coupled system? | [5+5] |
| | **OR** | |
| 3.a) | Explain different operations performed by the operating system. | |
| b) | What are the major activities of an operating system with regard to file management? Explain them briefly with their supporting system calls. | [5+5] |
| 4.a) | What is a process? Discuss various states of a process with the help of a process state transition diagram. | |
| b) | Distinguish between CPU bounded and I/O bounded processes. | [5+5] |
| | **OR** | |
| 5. | What is Semaphore? Explain producer consumer problem using semaphore. | [10] |
| 6.a) | Explain about Paging. | |
| b) | What is virtual memory? Discuss the benefits of virtual memory techniques. | [5+5] |
| | **OR** | |
| 7. | Discuss about page replacement algorithms with example. | [10] |

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8.a) Explain the issues in designing a file system.
b) Explain about the file system mounting. [5+5]

**OR**

9. Explain about the disk scheduling algorithms. [10]

10. Explain deadlock detection with suitable example. [10]

**OR**

11.a) Explain about Deadlock Avoidance.
b) Explain how to recovery from deadlock. [5+5]

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